

QUT-DV25: Run Procedure for Model Training and Evaluation

1. Prerequisites

1.1 System Requirements

- OS: Linux (Ubuntu 20.04+ recommended) or Windows with WSL2.
- Disk: 50 GB free space.
- RAM: Minimum 8 GB.
- GPU: Required for deep learning (CUDA-enabled).

1.2 Software Dependencies

Install the following tools:

```
sudo apt update
sudo apt install -y python3 python3-pip git gcc make
```

Ensure the following versions:

- Python: 3.8 or later.
- pip: 21.0 or later.

1.3 Python Libraries

Install all required libraries: `pip install -r requirements.txt`

requirements.txt:

```
numpy
pandas
matplotlib
scikit-learn
torch
torchvision
transformers
networkx
jupyter
notebook
tqdm
requests
beautifulsoup4
```

2. Folder structure (under Technical Validation and Benchmarks):

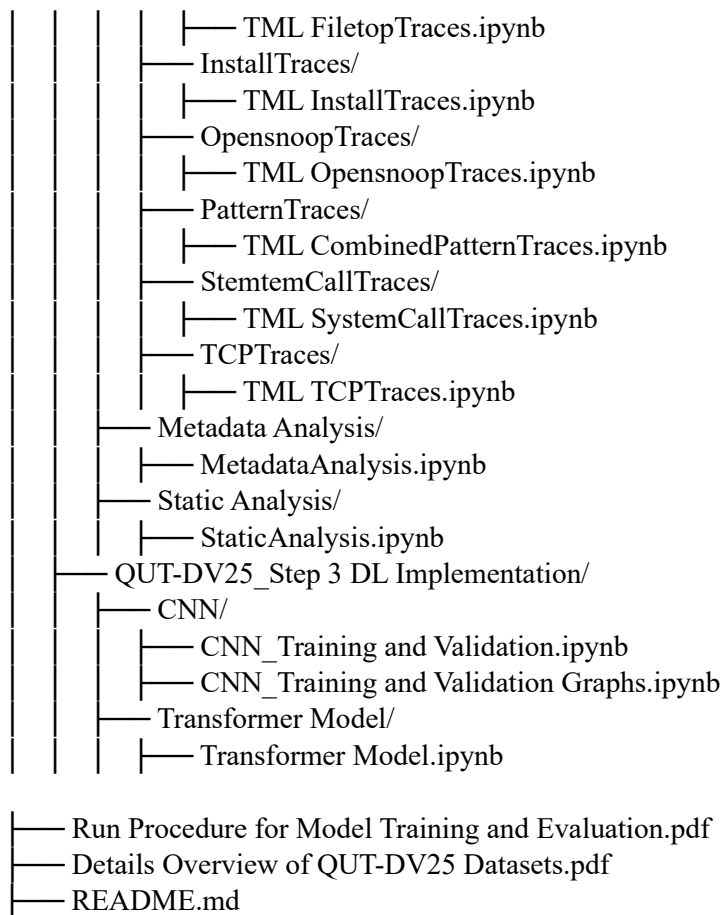
```
QUT-DV25/
├── Phase (i) Dataset Collection/
│   └── QUT-DV25_Step 1 Malicious Package Info/
│       ├── MaliciousPackageNameAndVersion.ipynb
│       └── Analysis_MaliciousPackageNameAndVersion.ipynb
```

- QUT-DV25_Step 2 Malicious Packages Info Check From Different Sources/
 - MaliciousPackagesDetailsFromDifferentWebsites.ipynb
 - Analysis_MaliciousPackagesDetailsFromDifferentWebsites.ipynb
- QUT-DV25_Step 3 Similarity Algorithms Implementation/
 - FileAnalysis.ipynb
 - runSimilarityChecking.sh
 - SimilarityCheck.py
 - Analysis_SimilarityAlgorithms-SearchBenignPackages.ipynb
- QUT-DV25_Step 4 Counterpart Benign Packages/
 - CounterpartBenignPackages.ipynb
 - DownloadSelectedBenignPackages.ipynb
 - FinalSelectedBenignPackages.ipynb
 - Analysis_CounterpartBenignPackages.ipynb

- Phase (ii) Dataset Labeling and Validation/
 - QUT-DV25_Step 1 Malicious Validation Report and Labeling/
 - MaliciousValidatorHPC.ipynb
 - run.sh
 - validation.py
 - Analysis_MaliciousValidator.ipynb
 - QUT-DV25_Step 2 Benign Validation Report and Labeling/
 - BenignValidationHPC.ipynb
 - benign_run.sh
 - benign_validation.py
 - Analysis_BenignValidator.ipynb
 - QUT-DV25_Step 3 Malicious Packages Metadata and Static Dataset/
 - MaliciousPackagesDetailsFromFiles.ipynb
 - Analysis_MaliciousPackagesDetailsFromFiles.ipynb

- Phase (iii) Trace Extraction/
 - QUT-DV25_Step 1 Isolated Env with RPIs/
 - Environment Setup with eBPF Program and Linux Guidelines
 - QUT-DV25_Step 2 Malicious Packages Traces/
 - prerequisites.sh
 - trace.sh
 - monitor.sh
 - QUT-DV25_Step 3 Benign Packages Traces/
 - prerequisites.sh
 - trace.sh
 - monitor.sh

- Technical Validation and Benchmarks/
 - QUT-DV25_Step 1 Traces Preprocessing-Feature Engineering/
 - Preprocessing-0.ipynb
 - Preprocessing-1.ipynb
 - Preprocessing-2.ipynb
 - QUT-DV25_Step 2 Traditional ML Implementation/
 - Dynamic Analysis/
 - CombinedTraces/
 - TML_CombinedTraces.ipynb
 - FiletopTraces/



3. Run Procedure by Phase

3.1 Phase (i) Dataset Collection

Step 1: Malicious Package Info

```

cd QUT-DV25/Phase (i) Dataset Collection/QUT-DV25_Step 1 Malicious Package Info
python -m notebook MaliciousPackagesNameAndVersion.ipynb
python -m notebook Analysis_MaliciousPackagesNameAndVersion.ipynb

```

Step 2: Check Info from Sources

```

cd ../QUT-DV25_Step 2 Malicious Packages Info Check From Different Sources
python -m notebook MaliciousPackagesDetailsFromDifferentWebsites.ipynb
python -m notebook AnalyMaliciousPackagesDetailsFromDifferentWebsites.ipynb

```

Step 3: Similarity Algorithms

```

cd ../QUT-DV25_Step 3 Similarity Algorithms Implementation
bash runSimilarityChecking.sh (HPC)
python SimilarityCheck.py
python -m notebook FileAnalysis.ipynb
python -m notebook Analysis_SimilarityAlgorithms-SearchBenignPackages.ipynb

```

Step 4: Counterpart Benign Packages

```
cd ../QUT-DV25_Step 4 Counterpart Benign Packages
python -m notebook CounterpartBenignPackages.ipynb
python -m notebook DownloadSelectedBenignPackages.ipynb
python -m notebook FinalSelectedBenignPackages.ipynb
python -m notebook Analysis_CounterpartBenignPackages.ipynb
```

3.2 Phase (ii) Dataset Labeling and Validation

Step 1: Malicious Validation

```
cd QUT-DV25/Phase (ii) Dataset Labeling and Validation/QUT-DV25_Step 1
Malicious Validation Report and Labeling
bash run.sh (HPC)
python validation.py
python -m notebook MaliciousValidatorHPC.ipynb
python -m notebook Analysis_MaliciousValidator.ipynb
```

Step 2: Benign Validation

```
cd ../QUT-DV25_Step 2 Benign Validation Report and Labeling
bash benign_run.sh (HPC)
python benign_validation.py
python -m notebook BenignValidatorHPC.ipynb
python -m notebook Analysis_BenignValidator.ipynb
```

Step 3: Static and Metadata Dataset

```
cd ../QUT-DV25_Step 3 Malicious Packages Metadata and Static Dataset
python -m notebook MaliciousPackagesDetailsFromFiles.ipynb
python -m notebook Analysis_MaliciousPackagesDetailsFromFiles.ipynb
```

3.3 Phase (iii) Trace Extraction

Note: Must be executed in a Linux-based isolated environment.

Step 1: Setup Environment

```
cd QUT-DV25/Phase (iii) Trace Extraction/QUT-DV25_Step 1 Isolated with RPIs
```

Also, follow manual setup instructions inside the guidelines file.

Step 2 & 3: Collect Traces

```
cd ../QUT-DV25_Step 2 Malicious Packages Traces
bash prerequisites.sh (HPC)
bash trace.sh (HPC)
bash monitor.sh (HPC)

cd ../../QUT-DV25_Step 3 Benign Packages Traces
```

```
bash prerequisites.sh (HPC)
bash trace.sh (HPC)
bash monitor.sh (HPC)
```

4. Technical Validation and Benchmarks

Step 1: Preprocessing and Feature Engineering

```
cd QUT-DV25/Technical Validation and Benchmarks/QUT-DV25_Step 1 Traces
Preprocessing-Feature Engineering
python -m notebook Preprocessing-0.ipynb
python -m notebook Preprocessing-1.ipynb
python -m notebook Preprocessing-2.ipynb
```

Step 2: Traditional ML Implementation

Dynamic Analysis:

```
cd ../QUT-DV25_Step 2 Traditional ML Implementation/ Dynamic Analysis/
CombinedTraces
python -m notebook TML_CombinedTraces.ipynb
```

Repeat similarly for:

- FiletopTraces/
- InstallTraces/
- OpensnoopTraces/
- PatternTraces/
- StemtemCallTraces/
- TCPTraces/

Metadata and Static Analysis:

```
cd ../../Metadata Analysis
python -m notebook MetadataAnalysis.ipynb
```

```
cd ../../Static Analysis
python -m notebook StaticAnalysis.ipynb
```

Step 3: DL Implementation

CNN Model:

```
cd ../../QUT-DV25_Step 3 DL Implementation/CNN
python -m notebook CNN_Training and Validation.ipynb
python -m notebook CNN_Training and Validation Graphs.ipynb
```

Transformer Model:

```
cd ../Transformer Model
python -m notebook Transformer Model.ipynb
```

5. Documentation

Refer to the following documents for more details:

- Run Procedure for Model Training and Evaluation.pdf
- Details Overview of QUT-DV25 Datasets.pdf
- README.md

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